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Brain Tumor Detection Using Segmentation Based Object Labelling Algorithm

E.Mallikarjuna [1],

[1,] Assistant Professor in Department of ECE, TEC Engineering College Tadipatri, AP. India.

. *Corresponding author: Mail Id: mallikarjuna.en@gmail.com

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ABSTRACT

In this paper we propose an efficient brain tumor detection method, which can detect tumor and locate it in the brain MRI images. This method exactly the tumor by using K-means algorithm followed by object labeling algorithm. Also some preprocessing steps (median filtering and morphological operations) are used for tumor detection purpose. It is observed that the experimental results of the proposed method gives better result in comparison to other technique. Brain tumor is inherently serious and life-threatening. Tumors are of different types and they have different Characteristics and different treatment. Tumor is an uncontrolled growth of tissues in any part of the body. This paper introduces inefficient detection of brain tumor from cerebral MRI images. Most Research in developed countries show that the number of people who have brain tumors were died due to the fact of inaccurate detection. CT scan or MRI that is directed into intracranial cavity produces a complete image to detect the tumor some preprocessing steps are used i.e. median filtering and morphological operation. After this by applying K means method then binarized image is labeled using Object Labeling algorithm. The purpose of this algorithm is to label different objects within the image. The tumor is extracted from the MR image and its exact position and the shape also determined. The stage of the tumor is displayed based on the amount of area calculated from the cluster.

Keywords: MRI images, CT scan, K means method and binarized image.

INTRODUCTION

This paper introduces an efficient detection of brain tumor from cerebral MRI images. To detect the tumor some preprocessing steps are used i.e. median filtering and morphological operation. After this by applying K means method then binarized image is labeled using Object Labeling algorithm. The purpose of this algorithm is to label different objects within the image. The tumor is extracted from the MR image and its exact position and the shape also determined. The stage of the tumor is displayed based on the amount of area calculated from the cluster.Digital image processing, the manipulation of images by computer, is relatively recent development in terms of man's ancient fascination with visual stimuli. In its short history, it has been applied to practically every type of images with varying degree of success. The inherent subjective appeal of pictorial displays attracts perhaps a disproportionate amount of attention from the scientists and also from the layman. Digital image processing like other glamour fields, suffers from myths, miss-connect ions, miss-understandings and miss-information. It is vast umbrella under which fall diverse aspect of optics, electronics, mathematics, photography graphics and computer technology. It is truly multidisciplinary endeavor ploughed with imprecise jargon.Several factor combine to indicate a lively future for digital image processing. A major factor is the declining cost of computer equipment. Several new technological trends promise to further promote digital image processing. These include parallel processing mode practical by low cost microprocessors, and the use of charge coupled devices (CCDs) for digitizing, storage during processing and display and large low cost of image storage arrays.

BRAIN ANATOMY OVERVIEW: The human brain which functions as the center for the control of all the parts of human body is a highly specialized organ that allows a human being to adapt and endure varying environmental conditions. The human brain enables a human to articulate words, execute actions, and share thoughts and feelings. In this section the tissue structure and anatomical parts of the brain are described to understand the purpose of this study. The brain is composed of two tissue types, namely gray matter (GM) and white matter (WM). Gray matter is made of neuronal and glial cells, also known as neuroglia or glia that controls brain activity and the basal nuclei which are the gray matter nuclei located deep within the white matter. The basal nuclei include: caudate nucleus, putamen, pallidum and claustrum. White matter fibers consist of many eliminated axons which connect the cerebral cortex with other brain regions. The left and the right hemispheres of the brain are connected by corpus callosum which is a thick band of white matter fibers. The brain also contains cerebrospinal fluid (CSF) which consists of glucose, salts, enzymes, and white blood cells. This fluid circulates through channels (ventricles) around the brain and the spinal cord to protect them from injury. There is also another tissue called meninges which are the membrane covering the brain and spinal cord.

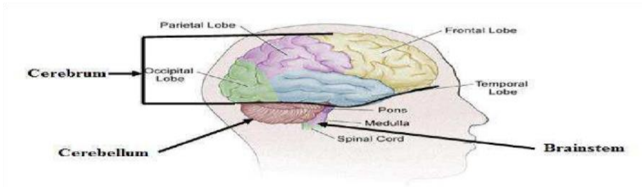

Fig 1: The Major Subdivision of Human

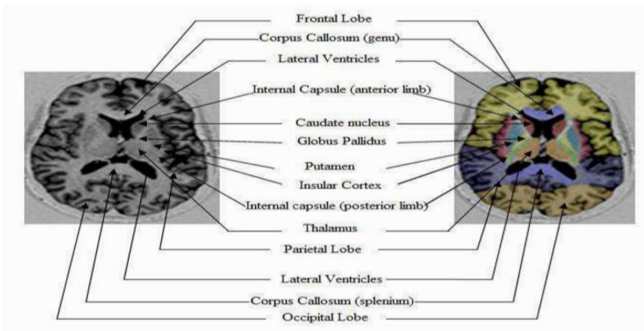

Figure 2: Overview Structure of Human Brain

Figure shows the anatomy of the brain. It is composed of the cerebrum and the brain stem. The cerebrum occupies the largest part of the brain. It is connected with the conscious thoughts, movement and sensations. It further consists of two halves, the right and the left hemispheres. Each controls the opposite side of the body. Moreover, each hemisphere is divided into four lobes: the frontal, temporal, parietal and occipital lobes. The cerebellum is the second largest structure of brain. It is connected with controlling motor functions of body such as walking, balance, posture and the general motor coordination. It is situated toward the back side of the brain and is linked to brain stems. Both, cerebellum and cerebrum have a very thin outer cortex of gray matter, internal white matter and small but deeply situated masses of the gray matter. The spinal cord is connected to the brainstem. It is located toward the bottom of the brain. Brainstem controls vital functions in human body such as motor, sensory pathways, cardiac, repository and reflexes. Brainstem has three structures: the midbrain, pons and medulla oblongata. Brain tumor is an abnormal growth of cells inside the skull. Normally the tumor will grow from the cells of the brain, blood vessels, nerves that emerge from the brain. There are two types of tumor which are benign (non-cancerous) and malignant (cancerous) tumors. The former is described as slow growing tumors that will exert potentially damaging pressure but it will not spread into surrounding brain tissue. However, the latter is described as rapid growing tumor and it is able to spread into surrounding brain. Tumors can damage the normal brain cells by producing inflammation, exerting pressure on parts of brain and increasing pressure within the skull.

Primary (true) brain tumors are commonly located in the posterior cranial fossa in children and in the anterior two thirds of the cerebral hemispheres in adults, although they can affect any part of the brain. The following are the most common symptoms of a brain tumor. However, each person may experience symptoms differently. Symptoms vary depending on the size and location of tumor.

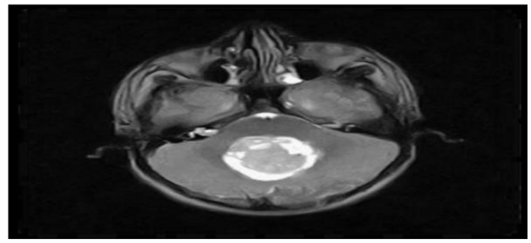

Fig 3: Presence of brain tumor

❖ Headache
❖ Vomiting (usually in the morning)
❖ nausea
❖ Personality changes
❖ Irritability
❖ Drowsiness
❖ Depression
❖ Decreased cardiac and respiratory function and, eventually, coma if not treated.

TUMOR: The word tumor is a synonym for a word neoplasm which is formed by an abnormal growth of cells Tumor 2.is something totally different from cancer. *C. Types* of Tumor. There are three common types of tumor:1. Benign; 2. Malignant (cancer can only be malignant).

1. Benign Tumor: A benign tumor is a tumor is the one that does not expand in an abrupt way; it doesn't affect its neighboring healthy tissues and also does not expand to non-adjacent tissues. Moles are the common example of benign tumors. **2. Malignant Tumor***:* Malignancy is the type of tumor that grows worse with the passage of time and ultimately results in the death of a person. Malignant is basically a medical term that describes a severe progressing disease. Malignant tumor is a term which is typically used for the description of cancer. D. Magnetic Resonance Imaging **MRI:** MRI is basically used in the biomedical to detect and visualize finer details in the internal structure of the body. This technique is basically used to detect the differences in the tissues which have a far better technique as compared to computed tomography (CT). So this makes this technique a very special one for the brain tumor detection and cancer imaging. CT uses ionizing radiation but MRI uses strong magnetic field to align the nuclear magnetization then radio frequencies changes the alignment of the magnetization which can be detected by the scanner. That signal can be further processed to create the extra information of the body.

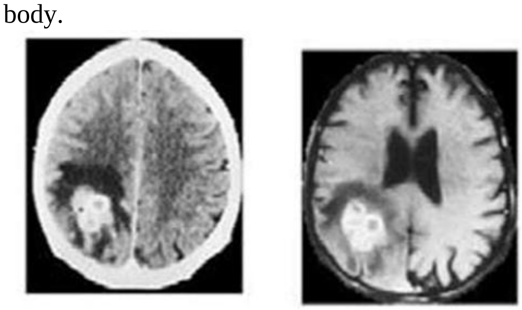

Fig: a **Fig: b**

Fig.4 : (a)CT scan Image and (b)MR Image

MRI and CT Analysis: Fused images from CT and MRI imagers are used for detection of tumor. The fused images are obtained from multiple modality images like Computed Tomography (CT) and Magnetic Resonance Image (MRI) as shown in Fig. (a) and (b). These multiple modality images play a key role in medical image processing; CT images which are used to ascertain the difference in tissue density and MRI provide an excellent contrast between various tissues of the body. CT images signify the difference in tissue density depending upon the tissues ability to react the X-rays, while MRI images provide contrast between different soft tissues. The above features make CT and MRI more suitable for the detection of tumor. The complementary and redundant information of both the source images are retained in the fused image, these information including the tumor size and location, which enable better detection of tumor, when compared to the source images.

BLOCK DIAGRAM:

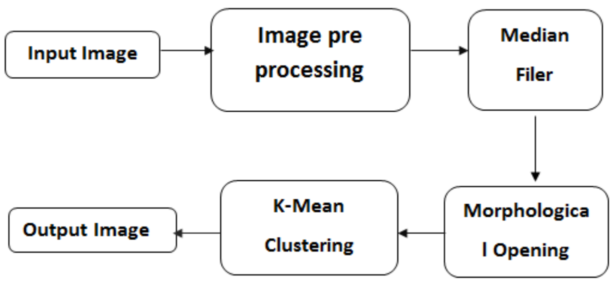

Fig 5 : Block diagram

Segmentation of Brain Tumors on Contrast Enhanced MR Images for Radiosurgery Applications , present a fast and robust practical tool for segmentation of solid tumors with minimal user interaction to assist clinicians and researchers in radio surgery planning and assessment of the response to the therapy. Particularly, a cellular automata (CA) based seeded tumor segmentation method on contrast enhanced T1 weighted magnetic resonance (MR) images, which standardizes the volume of interest (VOI) and seed selection, is proposed. Automatic segmentation of brain tumors in magnetic resonance images, presents an automatic method for segmentation of brain tumors in MRI. We use images of patients with glioblastoma multiform tumors. After pre-processing and removal of the regions that do not have useful information (e.g., eyes and scalp), we create a projection image for determining the primary location of the tumor. This image provides an overall view of the tumor. Then, we grow the primary region to segment the entire tumor. This method is automatic and independent of the operator. It segments low contrast tumors without requiring their exacta tissue boundaries. Brain Tumor Detection Using K-Means Clustering Segmentation, propose a segmentation method that uses the K-means clustering technique to track tumor objects in magnetic resonance (MR) brain images. The key concept in this color-based segmentation algorithm with K-means is to convert a given gray-level MR image into a color space image and then separate the position of tumor objects from other items of an MR image by using K-means clustering and histogram clustering. Brain Tumor Detection from Pre-Processed MRImages using Segmentation Techniques, has been proposed which can be successfully used to detect the brain tumor. This helps in determining the size and location of tumor. Edge based technique and color based segmentation are used. Edge-based segmentation has been implemented using operators e.g. Sobel, Prewitt, Canny and Laplacian of Gaussian operators. The color-based segmentation method has been accomplished using K-means clustering algorithm. The developed algorithm shows better result than Canny based edge detection. Brain MRI Segmentation for Tumor Detection using Cohesion based Self Merging Algorithm presents an algorithm for brain MRI segmentation to detect the location of the tumor. Segmentation using CSM based partitioned K-means clustering algorithm is applied. It is a self-merging algorithm. The noise effect is less. This approach is simple and less complex in computation. The computation time is also very less. In Multi fractal Texture Estimation for Detection and Segmentation of Brain Tumors, A stochastic model for characterizing tumor texture in brain magnetic resonance (MR) images is proposed. The efficacy of the model is demonstrated in patient independent brain tumor texture feature extraction and tumor segmentation in magnetic resonance images (MRIs). Due to complex appearance in MRI, brain tumor texture is formulated using a multi resolution-fractal model known as multi fractional Brownian motion. Detailed mathematical derivation for mBm model and corresponding novel algorithm to extract spatially varying multi fractal features are proposed. A multi fractal feature-based brain tumor segmentation method is developed next. To evaluate efficacy, tumor segmentation performance using proposed multi fractional feature is compared with that using Gabor-like multi scale text on feature. A New Approach to Image Segmentation for Brain Tumor detection using Pillar K-means Algorithm, presents a new approach to image segmentation using Pillar K-means algorithm. This segmentation method includes a new mechanism for grouping the elements of high resolution images in order to improve accuracy and reduce the computation time. The system uses K-means for image segmentation optimized by the algorithm after Pillar. The Pillar algorithm considers the placement of pillars should be located as far from each other to resist the pressure distribution of a roof, as same as the number of centroids between the data distribution. This algorithm is able to optimize the K-means clustering for image segmentation in the aspects of accuracy and computation time. This algorithm distributes all initial centroids according to the maximum cumulative distance metric.

HARD WARE REQUIRMENTS: The hardware for this project are

- ❖ Personal computer
- ❖ Key board
- ❖ Mouse
- ❖ OR a laptop

SOFTWARE REQUIRMENTS: MATLAB

Introduction to MATLAB

MATLAB is a high-performance language for technical computing. It integrates computation, visualization, and programming in an easy-to-use environment where problems and solutions are expressed in familiar mathematical notation. Typical uses include

- ❖ Math and computation
- ❖ Algorithm development
- ❖ Data acquisition
- ❖ Modeling, simulation, and prototyping
- ❖ Data analysis, exploration, and visualization
- ❖ Scientific and engineering graphics
- ❖ Application development, including graphical user interface building

RESULTS: The result for this project can be explained as follows. And the detection of brain tumor can be done by using k-means clustering.

- ❖ In this firstly we have to load the code into the MATLAB software. By running that code a window can be displayed as follows.
- ❖ Now by clicking the push button input image a window will be opened for the selection of the input image as shown below.

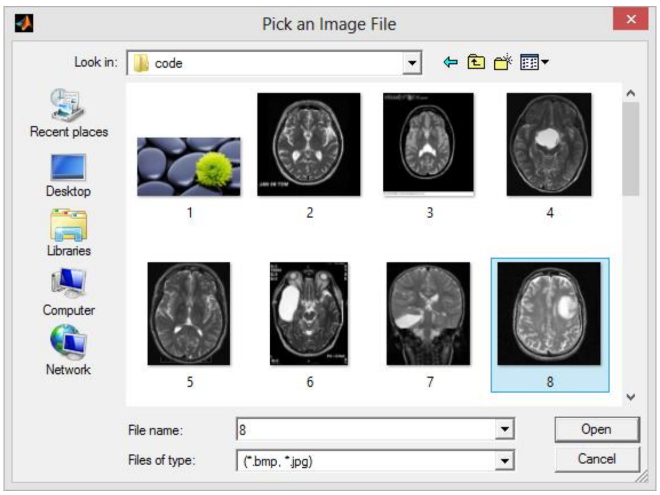

Fig 6: Input image selection window

- ❖ After selecting an image the image will be loaded.
- ❖ Now the image can under gone to preprocessing in this salt and pepper noise will we added to the image.
- ❖ Now we have to apply a median filter for removing the noise. Then the image
- ❖ after removing the noise .
- ❖ The next step the filtered image can under gone for the morphological operations. In this the image can be dilated i.e. blurred image can be formed.
- ❖ Now by clicking the K-mean image button a window will be displayed as follows. In that we have enter the no of clusters.
- ❖ Now the image can segmented into parts by this process all the skull part in the image will be removing by leaving the tumor part.

inally after the clustering process by clicking the push button named out image we may got the exact tumor part.

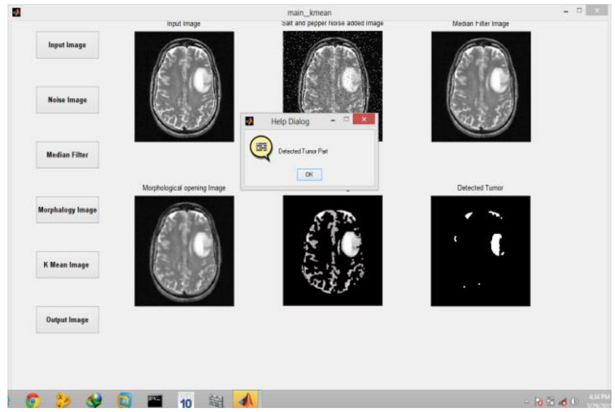

Fig 7: Window showing the output image

- ❖ After displaying the output image a window will be opened by clicking OK in that the output window will be cleared.

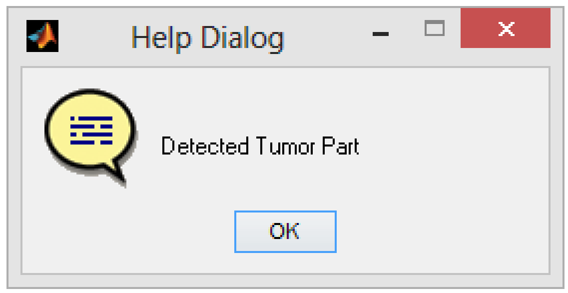

Fig 8: Window after deleting tumor part

CONCLUSION:

In this paper, brain tumor is detected from MRI images using K-means followed by Object Labeling algorithm. . This research was conducted to detect brain tumor using medical imaging techniques. The main technique used was segmentation, which is done using a method based segmentation, k-means clustering algorithm and morphological operators. The proposed segmentation method was experimented with MRI scanned images of human brains: thus locating tumor in the images. Samples of human brains were taken, scanned using MRI process and then were processed through segmentation methods thus giving efficient end results when the tumor is very close to bone, K-means and segmentation cannot segment the MRI image efficiently. So, new concept and different existing techniques can be used to improve the correctness of these algorithms

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